

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: CSE276

Course Title: ARTIFICIAL INTELLIGENCE
FOUNDATIONS

Course Outcome: CO2

Paper: Sample Multiple Choice Questions - ANSWER
KEY

Maximum Marks: 40 x 1 = 40

Unit: 2

Unit Title: Search Techniques, Knowledge
Representation and Reasoning

Unit Mapped RBT: L2 (Understand)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Identify the data structure that Depth First Search uses to hold its frontier.

- A) A stack <-- correct
- B) A queue
- C) A priority queue
- D) A hash table

Answer: A. A stack returns the most recently added node first, which is what makes the search plunge down one branch before considering any alternative.

Q2. State the evaluation function that A* search uses to order its frontier.

- A) $f(n) = g(n) - h(n)$
- B) $f(n) = g(n) + h(n)$ <-- correct
- C) $f(n) = g(n) \times h(n)$
- D) $f(n) = h(n) - g(n)$

Answer: B. A* adds the cost already incurred to the estimated cost remaining, which is what lets it be well informed and still optimal.

Q3. Identify the two parts of a production rule.

- A) A premise and a probability
- B) A slot and a default value
- C) A condition and an action <-- correct
- D) A node and a labelled link

Answer: C. A production rule fires its action whenever its condition is satisfied by the facts currently held in working memory.

Q4. State what the knowledge base of an expert system holds.

- A) The facts about the case in hand
- B) The program that applies the rules
- C) The record of the user's questions
- D) The rules supplied by the specialists <-- correct

Answer: D. Separating the rules from the engine that applies them is what allows the knowledge to be revised without rewriting the software.

Q5. Name the element of a semantic network that carries the relation between two concepts.

- A) A slot
- B) A labelled link <-- correct
- C) A production rule
- D) A clause

Answer: B. Concepts are drawn as nodes and the named relations between them as directed labelled links, which is what makes inheritance readable off the graph.

Q6. Identify the named compartments in which a frame holds the attributes of a concept.

- A) Links
- B) Clauses

- C) Premises
- D) Slots <-- correct

Answer: D. Each slot carries a value, a default or a constraint, and a frame may inherit the slots of a parent frame.

Q7. State the name of the logical connective written as an arrow between two propositions.

- A) Implication <-- correct
- B) Conjunction
- C) Disjunction
- D) Negation

Answer: A. P implies Q asserts that whenever P holds, Q holds too. It says nothing at all about the case in which P is false.

Q8. Name the quantifier of first-order logic that asserts a property holds of every object.

- A) The existential quantifier
- B) The negation operator
- C) The universal quantifier <-- correct
- D) The implication operator

Answer: C. The universal quantifier claims the predicate is true of every object in the domain; the existential claims only that at least one such object exists.

Q9. Identify the theorem used to revise a probability in the light of new evidence.

- A) Modus ponens
- B) De Morgan's law
- C) Bayes' theorem <-- correct
- D) The chain rule of logic

Answer: C. Bayes' theorem converts a prior belief into a posterior one by weighing how likely the observed evidence would be under each hypothesis.

Q10. State what the term $P(A)$ represents in Bayes' theorem.

- A) The prior probability of A <-- correct
- B) The posterior probability of A
- C) The likelihood of the evidence
- D) The probability of the evidence

Answer: A. The prior is the belief held before the evidence is seen. Bayes' theorem updates it into the posterior $P(A \text{ given } B)$.

Q11. Name the term $P(A \text{ given } B)$ in Bayes' theorem.

- A) The prior probability
- B) The likelihood
- C) The marginal probability
- D) The posterior probability <-- correct

Answer: D. The posterior is the revised belief in A once B has been observed, and it is the quantity the calculation sets out to obtain.

Q12. Identify the search that orders its frontier by the estimated remaining cost alone.

- A) Breadth First Search
- B) Best First Search <-- correct
- C) Depth First Search
- D) A* search

Answer: B. Greedy Best First Search uses $h(n)$ by itself. Ignoring the cost already spent is why it is neither complete nor optimal.

Q13. State the property of a heuristic that never overestimates the cost remaining to a goal.

- A) Completeness
- B) Optimality
- C) Monotonicity of cost
- D) Admissibility <-- correct

Answer: D. Admissibility is what guarantees that A* returns an optimal solution, because no optimal path can be discarded as too expensive.

Q14. Name the local search that moves to the best neighbouring state and halts when no neighbour is better.

- A) Breadth First Search
- B) Best First Search
- C) Hill Climbing <-- correct
- D) A* search

Answer: C. Hill Climbing keeps no search tree and no memory of where it has been, which is why it is cheap but easily trapped.

Q15. State the term for the store of facts an expert system accumulates about the case in hand.

- A) The knowledge base
- B) The working memory <-- correct
- C) The inference engine
- D) The user interface

Answer: B. The knowledge base holds the general rules; the working memory holds the particular facts, and the engine matches one against the other.

Q16. Name the search strategy that expands the shallowest unexpanded node first.

- A) Breadth First Search <-- correct
- B) Depth First Search
- C) Hill Climbing
- D) Best First Search

Answer: A. Expanding by depth means the first goal it reaches is the shallowest one, which is optimal whenever every step costs the same.

Q17. Explain why Depth First Search needs far less memory than Breadth First Search.

- A) It holds no nodes in memory at any time

- B) It holds only the current path and its siblings <-- correct
- C) It examines fewer nodes in total than BFS
- D) It stores only the nodes at the goal depth

Answer: B. Depth First keeps a single branch, so its memory grows with the depth rather than the width. Breadth First must hold an entire level at once.

Q18. Distinguish an admissible heuristic from one that is not.

- A) An admissible heuristic never exceeds the true remaining cost <-- correct
- B) An admissible heuristic never falls below the true cost
- C) An admissible heuristic returns the exact cost always
- D) An admissible heuristic returns the same value everywhere

Answer: A. Underestimating keeps A* optimistic about every unexplored path, so no optimal route is pruned before it can be examined.

Q19. Describe what Best First Search ignores that A* takes into account.

- A) The estimated cost remaining to the goal
- B) The number of successors each node has
- C) The depth of the node in the search tree
- D) The cost already incurred in reaching the node <-- correct

Answer: D. Greedy search follows the most promising-looking direction without regard to what has been spent, which is how it can commit to an expensive route.

Q20. Interpret the behaviour of A* when its heuristic returns the exact remaining cost at every node.

- A) It expands every node in the whole space
- B) It behaves exactly like Depth First Search
- C) It expands only the nodes on an optimal path <-- correct
- D) It fails to terminate on any problem at all

Answer: C. A perfect heuristic makes $f(n)$ constant along an optimal path and larger elsewhere, so the search walks straight to the goal. Such a heuristic is rarely available.

Q21. Predict the behaviour of Hill Climbing when it reaches a plateau.

- A) It has no gradient to follow and stalls or wanders <-- correct
- B) It jumps directly to the global maximum
- C) It reverses and returns to the start state
- D) It reports that no solution can exist

Answer: A. On a plateau every neighbour scores the same, so the rule that selects the best neighbour gives the search no direction at all.

Q22. Recognize the representation described: concepts are drawn as nodes and the relations between them as labelled arrows.

- A) A frame system
- B) A production system
- C) A semantic network <-- correct
- D) A propositional clause

Answer: C. The graph form makes inheritance visible: a property attached to a general node is available to every node beneath it along the is-a links.

Q23. Explain what inheritance contributes to a frame system.

- A) A frame copies every fact in working memory
- B) A frame takes its parent's slots unless it overrides them <-- correct
- C) A frame acquires the rules of the inference engine
- D) A frame receives a probability from its parent

Answer: B. Shared properties are stated once on the general frame, and a specific frame need record only what differs, which keeps the knowledge base compact and consistent.

Q24. Summarize what an expert system offers that ordinary software does not.

- A) It runs considerably faster than ordinary software
- B) It requires no knowledge from any human specialist
- C) It improves its own rules from every case it sees
- D) It reasons from stated rules and can explain its conclusion <-- correct

Answer: D. The knowledge is declarative and inspectable, so the chain of rules that produced an answer can be shown. It does not, however, learn from its mistakes.

Q25. Distinguish a constant from a variable in first-order predicate logic.

- A) A variable names one object, a constant ranges over many
- B) Both of the two name exactly one object each
- C) A constant names one object, a variable ranges over many <-- correct
- D) Both of the two range over many objects

Answer: C. Student(rahul) makes a claim about one person; for all x , Student(x) implies Enrolled(x) makes a claim about every object the variable can stand for.

Q26. Explain why A* is described as an informed search.

- A) It uses the labels attached to the training data
- B) It examines every node before choosing one
- C) It requires the solution depth to be known

D) It uses knowledge of the goal through its heuristic
<-- correct

Answer: D. Uninformed methods know only the structure of the problem. A heuristic carries problem-specific knowledge about how far a state is from the goal.

Q27. Interpret a Bayes calculation in which the prior is very small while the likelihood of the evidence is high.

- A) The posterior can still remain low <-- correct
- B) The posterior must become close to one
- C) The posterior is unaffected by the prior
- D) The calculation cannot be carried out

Answer: A. A rare condition stays rare after a positive test unless the test is extremely specific. Ignoring the small prior is the base-rate fallacy.

Q28. Predict the effect on A* of a heuristic that returns a very large value at every node.

- A) It behaves like Breadth First Search exactly
- B) It behaves like a greedy search and may lose optimality <-- correct
- C) It refuses to expand any node at all
- D) It is guaranteed to find the cheapest path

Answer: B. When h dominates g the sum is decided almost entirely by the estimate, and a heuristic that overestimates can push the true best path out of contention.

Q29. Describe the role of the working memory in a production system.

- A) It holds the rules supplied by the specialists
- B) It holds the program that matches the rules
- C) It holds the questions the user has asked
- D) It holds the facts known about the current case <-- correct

Answer: D. Rules are matched against working memory; when one fires, its action adds or alters facts there, and the cycle repeats until nothing new is produced.

Q30. Recognize the representation described: a record holding the attributes of one concept in named compartments, with defaults taken from a parent.

- A) A semantic network
- B) A frame <-- correct
- C) A production rule
- D) A predicate sentence

Answer: B. Frames suit the structured description of a single object, including its defaults and permitted value ranges, which a network of binary links represents only awkwardly.

Q31. Explain why a heuristic function must be inexpensive to compute.

- A) It must be recomputed after every solution

- B) It has to be stored alongside every state
- C) It is evaluated at every node the search considers <-- correct

D) It is required to be perfectly accurate

Answer: C. A more accurate heuristic expands fewer nodes but costs more per node. The useful heuristic is the one that wins that exchange overall.

Q32. Distinguish a fact from a rule in an expert system.

- A) A fact states a particular, a rule states a general relation <-- correct
- B) A rule states a particular, a fact a general relation
- C) Both of the two state particular observations
- D) Both of the two state general relations

Answer: A. The rules make up the knowledge base and apply to any case; the facts describe the case actually in front of the system.

Q33. Explain what makes Hill Climbing a local search rather than a systematic one.

- A) It examines only the neighbours of its current state <-- correct
- B) It examines every state in the whole space
- C) It keeps a complete record of its search tree
- D) It requires an admissible heuristic to run

Answer: A. There is no frontier and no memory of alternatives, so the search cannot back up to a promising branch it passed earlier.

Q34. Interpret the sentence: for all x , Student(x) implies Enrolled(x).

- A) Some student is enrolled
- B) No student is enrolled
- C) Exactly one student is enrolled
- D) Every student is enrolled <-- correct

Answer: D. The universal quantifier applies the implication to every object, so the sentence claims that being a student is sufficient for being enrolled.

Q35. Predict the result of applying Breadth First Search to a tree with a very large branching factor.

- A) The goal is reached in a constant number of steps
 - B) Memory is exhausted long before the goal is reached <-- correct
 - C) The search silently switches to Depth First
 - D) The search returns a suboptimal solution quickly
- Answer: B.** The frontier holds an entire level, and each level is b times larger than the one before, so the memory requirement grows exponentially with depth.

Q36. Summarize the applications for which the search techniques of this unit are used.

- A) Training deep neural networks on images
- B) Compressing files for efficient storage
- C) Route planning, puzzle solving and scheduling tasks <-- correct

D) Encrypting messages sent over a network

Answer: C. Any problem posed as states, actions and a goal test is amenable to search: navigation, the eight-puzzle, timetabling and resource allocation among them.

Q37. Describe what the likelihood term contributes to Bayes' theorem.

- A) How probable the hypothesis is before evidence
- B) How probable the evidence is over all hypotheses
- C) How probable the evidence is if the hypothesis holds <-- correct
- D) How probable the hypothesis is after the evidence

Answer: C. The likelihood $P(B \text{ given } A)$ is what distinguishes between hypotheses: evidence that is far more probable under one of them shifts belief towards it.

Q38. Explain why the knowledge base of an expert system is kept separate from its inference engine.

- A) The engine then runs considerably faster
- B) The knowledge can be revised without rewriting the program <-- correct
- C) The rules can then be hidden from the user
- D) The system then needs no working memory

Answer: B. A specialist can add or correct rules without touching the reasoning machinery, and the

same engine can serve several different knowledge bases.

Q39. Distinguish a semantic network from a frame system by the way each holds the properties of one concept.

- A) A frame spreads them over links, a network gathers them
- B) Both of the two gather the properties into slots
- C) Both of the two spread the properties over links
- D) A network spreads them over links, a frame gathers them in slots <-- correct

Answer: D. A frame keeps everything about one concept in a single structure, including defaults and constraints. A network expresses each property as a separate binary link.

Q40. Explain why probability is used to reason when the available evidence is incomplete.

- A) It expresses degrees of belief that logic cannot <-- correct
- B) It removes the need for any evidence at all
- C) It guarantees that the conclusion is correct
- D) It converts every rule into a certain fact

Answer: A. Strict logic must call a sentence true or false. Probability lets a conclusion be held with a stated degree of confidence and revised as evidence arrives.

----- End of Sample Question Set -----